

STCLOUD 05.08

Virtualization and Cloud Computing

There are three types of deployments - traditional (bare metal), virtualization (still private cloud), Cloud (either public or private)

Traditional

- physical servers and datacenters to deliver computing and other IT services
- used to be a practice dati
- Mostly telcos
 - Server tells you when you've done your planned number of calls
- Host computer would typically be installed with a single OS directly sa hardware. Also called Blackbox. Lahat ng proprietary nila.
- Possible issues: resource allocation, difficulty in scaling.
 - Kung gusto ko magupgrade, madidispore yung mga lumang hardware. So essentially, its costly.



Image 1: Traditional / Physical server

In the physical server, install an OS then you can install a specific application to do what it needs to do. Ang problema it only does ONE specific function, so like One App in One Server. If the demand of that app goes down, it becomes under-utilized, and then masasayang kung magkano yung binayad mo for that server

Let's say magnitude is growing in data center. if mga 5 to 15% lang ginagamit, its a huge waste of IT resources.

Virtualization

- is a technology (Cloud is a service)
- create virtual representations of servers, storage and networks of other physical machines and resources
- Virtual Software mimics the functions of physical hardware
- So we use a lot of Virtual Machines. Three applications that are Virtualized now share the same hardware.

How to achieve virtualization

We dynamically allocate/deallocate VMs

VM Profile - CPU, Memory, Storage, Network Interfaces.

So we install then a Hypervisor which manages Virtual Machines which now share the same physical servers. Hypervisors allow multiple virtual machines to run on a single physical server

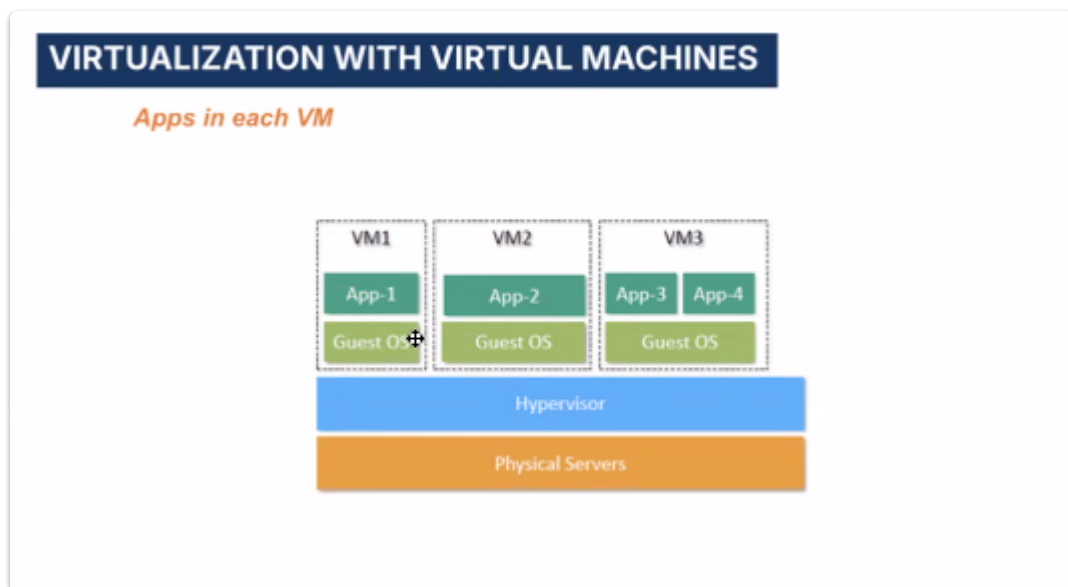


Image 2: We get to do the same things as traditional with less cost

One advantage of Virtualized, is that we can take what we call a Snapshot. In traditional, pag may mangyari sa server, there is a huge risk na mawawala lahat ng data sa server. Pero in virtualized, we can take a Snapshot to create a VMI (VM Image) which if you install ulit sa hypervisor, it will do the same thing with the same data (an exact replica) of the virtual machine.

- Virtual Machine - self contained, independent, isolated software container with an operating system and an application inside.
- Multiple VMs on a single computer results in several OS and applications to run on one physical server or host.

Buying a phone, you dont buy a phone for each possible need. You buy a phone, a single phone, that can do all requirements you need.

Without Virtualization, huge inefficiencies and excessive operating costs

With virtualization, economies of scale and greater efficiency

Cloud Computing

- On demand delivery of IT resources. Virtualization is a tech, Cloud is a service which is that first sentence.
- The tech of virtualization enables the services that Cloud can offer
- You can access tech services such as computing power, storage and databases, on as-needed basis from a provider like Amazon Web Services
- Cloud computing is a general term for anything that involves delivering hosted services over the internet.
- The goal of cloud computing is to provide easy, scalable access to computing resources and IT services.

Benefits

1. On-demand like Netflix

- Cloud services are on-demand which means you can access the service anywhere and anytime

2. Cheaper

- You don't need to purchase computers, rent office space or hire staff. Various models of cloud services offer one or all of those options. Meaning huge savings on company

3. It's Flexible

- Since you only pay for what you want and what you want is accessible anywhere, you can increase or decrease the scope of projects as needed

4. Mobility

- Storing information in cloud means that users can access it from anywhere

5. Disaster Recovery

- Storing data in cloud guarantees that users can always access their data even if devices are inoperable.
- With cloud based services, organizations can quickly recover their data in the event of emergencies such as natural disasters or power outages.

Cloud Computing vs Traditional Web Hosting

1. Users can access large amounts of computing power on demand

1. You can change the computing power to accommodate depending on your needs

2. Elastic -> user can have as much or as little of a service any given time

3. Service is fully managed by provider. Significant innovations in virtualization and distributed computing as well as improved access to high-speed internet have accelerated interest in cloud computing

Types of Cloud Computing

IaaS, PaaS, SaaS

Pizza as a Service

Pizza as a Service

On-Premises – like a homemade pizza, made from scratch, you do everything yourself (no change so far). Example: Datacentre

Infrastructure as a Service – You share a kitchen with others. The utilities and oven are provided, but you make and cook the pizza yourself. Example: EC2

Platform as a Service – You order a pizza for collection, the pizzeria make and cook the pizza using their facilities. Example: App Engine

Software as a Service – You go to someone's house for a party, they provide the pizza and invite others round for you to meet. Conversation with the guests is still your responsibility! Example: Gmail

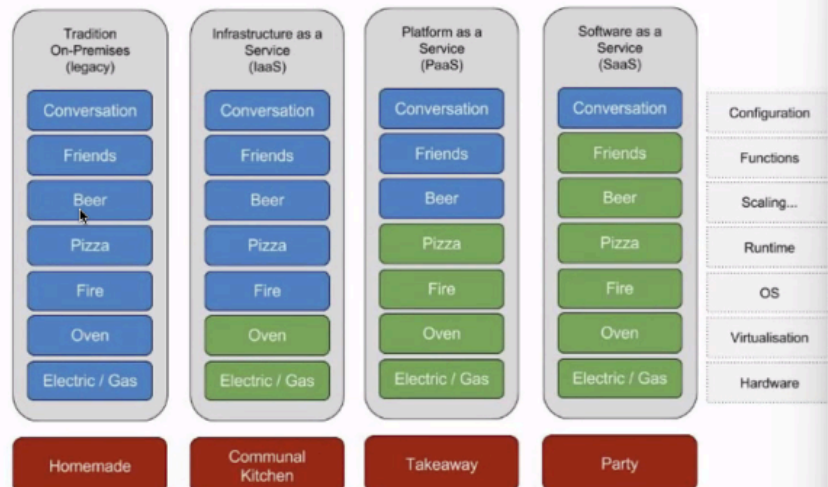


Image 3: Pizza as a service

A disadvantage of SaaS, in terms of customization, may be that it is difficult to customize

Types of Cloud Computing

	Key Benefits	Challenges	Examples
Infrastructure as a Service	- Scalability - Cost & Time Savings - Flexibility	- Integration - Security - Staff Training	  
Platform as a Service	- Speed - Customization - Flexibility	- Integration - Compatibility - Control	  
Software as a Service	- All Inclusive - Cost Efficient - Accessible	- Customization - Interoperability - Control	  

Reference: [SaaS vs PaaS vs IaaS: What's the Difference & How To Choose \(thinklogic.com\)](#)

Image 4: Chart of examples

Cloud Deployment Models

The deployment model is defined according to where the infrastructure for the deployment resides and who has control over that infrastructure.

Basically, the decision depends on the value propositions and different costs associated with it.

The consideration usually is ano mas madali ano mas cheaper

Public Cloud

- Sells services to anyone on the internet.
- This type of cloud deployment model supports all users who want to make use of a computing resources such as hardware or software on a subscription basis
- Di mo talaga owned yung mga servers.
- Most common uses are for application development, file sharing, email service

Private Cloud

- Sayo ito
- Operated solely for the organization.
- Infrastructure will only host a company's applications.
- So kanila mga location, kanila data centers.
- More expensive than public clouds due to the capital expenditure involved in acquiring and maintaining them.
- But better able to address security and privacy concerns

Hybrid Cloud

- Interconnected private and public cloud infrastructure.
- Make use of this model ****

Cloud Deployment Models Comparison

	Public Cloud	Private Cloud	Community Cloud	Hybrid Cloud
Set Up and Use	Handle In-house	Requires IT Professionals		
Privacy and Security	Low	High	Medium	Varies from Low to High
Control of Data	Low	High	Medium	Medium
Overall Reliability	Medium	High	Medium	Medium to High
Flexibility and Scalability	High	Medium (Stable Capacity)	Medium (Stable Capacity)	Very High
Cost	Lowest	Relatively High	Variable	Medium/Variable
Hardware	Third-Party	Variable (can be on-site or third party)	Variable	Medium/Variable

Image 5: Cloud deployment charts

Cloud Computing is on demand delivery of IT resources

Three basic types SaaS, IaaS, PaaS

Public, Private, Community and Hybrid deployment models